Evaluate yourself 3:

a. Write the IUPAC name for the following compounds.

(i) $K_2[Fe(CN)_3(Cl)_2(NH_3)]$

(ii) $[Cr(CN)_2(H_2O)_4][Co(ox)_2(en)]$

(iii) $[Cu(NH_3)_2Cl_2]$

(iv) $[Cr(NH_3)_3(NC)_2(H_2O)]^+$

(v) $[Fe(CN)_6]^{4-}$

b. Give the structure for the following compounds.

(i) Diamminesilver(I) dicyanidoargentate(I)

(ii) Pentaammine nitrito-κNcobalt (III) ion

(iii) Hexafluorido cobaltate (III) ion

(iv) Dichloridobis(ethylenediamine) Cobalt (IV) sulphate

(v) Tetracarbonylnickel (0)

5.5 Isomerism in coordination compounds

We have already learnt the concept of isomerism in the context of organic compounds, in the previous year chemistry classes. Similarly, coordination compounds also exhibit isomerism. Isomerism is the phenomenon in which more than one coordination compounds having the same molecular formula have different physical and chemical properties due to different arrangement of ligands around the central metal atom. The following flow chart gives an overview of the common types of isomerism observed in coordination compounds,

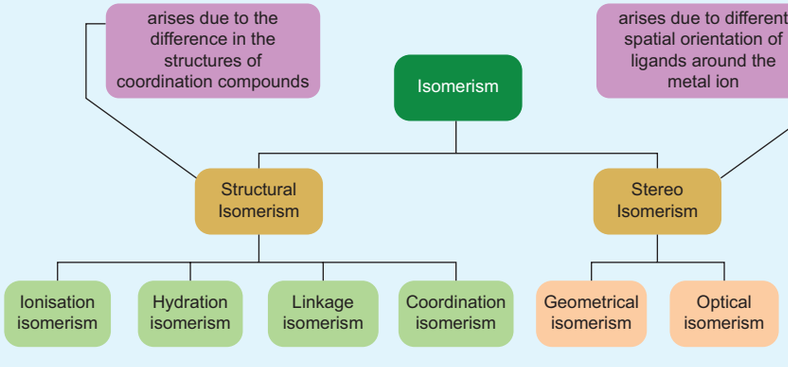


Figure 5.2 Isomerism in coordination compounds

5.5.1 Structural isomers

The coordination compounds with same formula, but have different connections among their constituent atoms are called structural isomers or constitutional isomers. Four common types of structural isomers are discussed below.

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